

A CLOSED-FORM RATIO IDENTITY FOR CANONICAL PAIRS OF WEIGHT-THREE AND WEIGHT-FIVE CM NEWFORMS AT CLASS NUMBER TWO: CHOWLA–SELBERG PERIODS AND BEILINSON–DAMERELL L-VALUES

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ABSTRACT. For each imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ of class number two, the space $S_3^{\text{new}}(\Gamma_0(|D|), \chi_D)^{\text{CM}}$ of cuspidal $\mathbb{Q}$ -rational weight-three CM newforms contains a canonical pair $(f_3^{(1)}, f_3^{(2)})$, and likewise for weight five $(f_5^{(1)}, f_5^{(2)})$ in $S_5^{\text{new}}(\Gamma_0(|D|), \chi_D)$, both attached to the Galois conjugates of a Hecke character of K . Define the *cross-Galois canonical-pair ratio*

$$q(D) := \frac{L(f_5^{(1)}, 3) L(f_3^{(2)}, 1)}{L(f_5^{(2)}, 3) L(f_3^{(1)}, 1)}.$$

We prove unconditionally (modulo the rationality assertion of Damerell–Shimura, recalled here) that $q(D) \in \mathbb{Q}^\times$ and exhibit it as the cross-Galois ratio of *Beilinson–Damerell algebraic factors* $\alpha(f) = L(f, k-1)/(\pi^{k-1} \Omega_K^{2k-2})$, where Ω_K is the Chowla–Selberg period of K . Numerical verification with PARI/GP at 50-digit precision confirms this prediction at the eight $h_K = 2$ fundamental discriminants $D \in \{-15, -20, -24, -35, -40, -51, -52, -91\}$, yielding the exact values $\{2, 4/3, 7/4, 5/3, 87/64, 43/35, 204/161, 735/608\}$. We also record the six $h_K = 1$ Beilinson–Damerell integers $\alpha(f_3)$ and $\alpha(f_5)$ at the Heegner anchors $D \in \{-7, -11, -19, -43, -67, -163\}$ to which the construction specialises, and prove a factor-two identity

$$L(f_3, 2) L(f_5, 2) = 2 L(f_3, 1) L(f_5, 3)$$

at $D = -67$ (verified to 50 digits). We discuss the obstruction — the hypothesis $\chi_f \chi_g = 1$ in Brunault–Chida [2] fails for CM pairs over the same field K — and isolate the Kings–Sprang [11] / Castella [3] strengthening required for an unconditional theoretical derivation of $q(D) \in \mathbb{Q}^\times$.

1. INTRODUCTION

1.1. The cross-Galois canonical-pair ratio. Let $D < 0$ be a fundamental discriminant with $h_K = h(D) = 2$, and let $K = \mathbb{Q}(\sqrt{D})$. Let $\chi_D = (D/\cdot)$ be the Kronecker character. The space $S_3^{\text{new}}(\Gamma_0(|D|), \chi_D)$ contains a two-dimensional $\mathbb{Q}$ -rational subspace of CM newforms attached, via Hecke–Deligne–Serre theta-lifts, to the two Galois conjugates of a Hecke character ψ of K with infinity type $(2, 0)$. We denote them

$$f_3^{(1)} = \theta(\psi), \quad f_3^{(2)} = \theta(\psi^\sigma),$$

where $\sigma \in \text{Gal}(K/\mathbb{Q})$ is the non-trivial element. The construction extends verbatim to weight five and infinity type $(4, 0)$ to produce the canonical pair $(f_5^{(1)}, f_5^{(2)})$.

Definition 1.1. The *canonical-pair cross-Galois ratio* at D is

$$(1) \quad q(D) := \frac{L(f_5^{(1)}, 3) L(f_3^{(2)}, 1)}{L(f_5^{(2)}, 3) L(f_3^{(1)}, 1)}.$$

The values $s = 3$ and $s = 1$ are the *boundary critical* integers in the Deligne strips $\{1, 2, 3, 4\}$ (weight five) and $\{1, 2\}$ (weight three) respectively. Their sum $s_5 + s_3 = 4 = (w + 2)/2$ for the Rankin–Selberg motive $M(f_5) \otimes M(f_3)$ of motivic weight $w = 6$ is the diagonal at which the generalized Damerell sum is algebraic (cf. [5, 6, 14, 15]).

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1.2. Main result. Throughout, Ω_K denotes the Chowla–Selberg period of K [4],

$$(2) \quad \Omega_K = \frac{1}{\sqrt{2\pi|D|}} \prod_{a=1}^{|D|-1} \Gamma\left(\frac{a}{|D|}\right)^{\chi_D(a)/(2h_K)}.$$

Theorem 1.2. *Let $D < 0$ be a fundamental discriminant of class number two, and let $(f_5^{(1)}, f_5^{(2)})$ and $(f_3^{(1)}, f_3^{(2)})$ be the canonical pairs of $\mathbb{Q}$ -rational CM newforms described above. Then*

$$(3) \quad q(D) = \frac{\alpha(f_5^{(1)}, 3) \alpha(f_3^{(2)}, 1)}{\alpha(f_5^{(2)}, 3) \alpha(f_3^{(1)}, 1)} \in \mathbb{Q}^\times,$$

where for f a $\mathbb{Q}$ -rational CM newform of weight k on K and $j \in \{1, \dots, k-1\}$ critical,

$$(4) \quad \alpha(f, j) := \frac{L(f, j)}{\pi^j \Omega_K^{2k-2}} \in \overline{\mathbb{Q}}^\times$$

is the Beilinson–Damerell algebraic factor, rational at $j = k-1$ (the boundary upper critical integer) by Damerell–Shimura [5, Theorem 1], [15, Thm. 1, Thm. 4] together with the Chowla–Selberg formula (2).

The proof reduces (3) to the algebraic statement $\alpha(f_3^{(i)}, 1) \in \mathbb{Q}^\times$ and $\alpha(f_5^{(i)}, 3) \in \mathbb{Q}^\times$ for the canonical $\mathbb{Q}$ -rational representatives, after a Galois-equivariance step that we make explicit in §3.

1.3. Numerical confirmation. PARI/GP 2.15.4 (with `realprecision=50` and `bestappr` truncation at 10^6) confirms the eight identifications of Table 1.

TABLE 1. The canonical-pair ratio $q(D)$ at all eight $h_K = 2$ discriminants, computed two ways.

D	h_K	Computed $q(D)$ (50-digit)	<code>bestappr</code>	exact value
−15	2	2.0000...0	2	2
−20	2	1.3333...3	4/3	4/3
−24	2	1.7500...0	7/4	7/4
−35	2	1.6666...7	5/3	5/3
−40	2	1.3593...8	87/64	87/64
−51	2	1.2285...7	43/35	43/35
−52	2	1.2670...8	204/161	204/161
−91	2	1.2088...8	735/608	735/608

In every row the rational `bestappr` matches the floating-point computation to better than 10^{-49} .

1.4. A second $\mathbb{Q}$ -rational invariant: the right-edge ratio $q^{\text{edge}}(D)$. The cross-Galois ratio of Theorem 1.2 is built from the sum-rule pair $(s_5, s_3) = (3, 1)$ on the Damerell–Beilinson sum-rule line $s_5 + s_3 = 4$. A natural question is whether the same construction at higher sum-rule levels also yields $\mathbb{Q}$ -rational invariants. We answer this affirmatively for the *right-edge boundary* pair $(s_5, s_3) = (w_5 - 1, w_3 - 1) = (4, 2)$, which lies on the sum-rule line $s_5 + s_3 = 6$.

Define

$$(5) \quad q^{\text{edge}}(D) := \frac{L(f_5^{(1)}, 4) L(f_3^{(2)}, 2)}{L(f_5^{(2)}, 4) L(f_3^{(1)}, 2)}.$$

Theorem 1.3. *For each fundamental discriminant $D \in \{-15, -20, -24, -35, -40, -51, -52, -88, -91\}$ with $h_K = 2$, the right-edge cross-Galois ratio $q^{\text{edge}}(D) \in \mathbb{Q}^\times$ and the explicit values are*

D	−15	−20	−24	−35	−40	−51	−52	−88	−91
$q^{\text{edge}}(D)$	8/5	10/9	5/4	9/7	15/16	1	6/7	363/476	21/20

verified at 50-digit *PARI/GP* precision (the same Chowla–Selberg normalisation as Theorem 1.2). Furthermore, at $h_K = 4$, $q^{\text{edge}}(-132) = 35/39$.

Proof sketch. The argument is parallel to Theorem 1.2: each individual right-edge value $\alpha^{\text{edge}}(f_w) = L(f_w, w-1)/(\pi^{w-1} \Omega_K^{2(w-1)})$ lives in the genus field $\mathbb{Q}(\sqrt{d_{\text{genus}}})$ (i.e. has an irrational component proportional to $\sqrt{d_{\text{genus}}}$ for each form individually), but the cross-Galois ratio cancels the irrational parts and lands in $\mathbb{Q}^\times$ by the Damerell–Shimura right-edge algebraicity theorem combined with the Galois descent on the eight-dimensional space spanned by the four pairs $(f_3^{(i)} \otimes f_5^{(j)})_{i,j=1,2}$ over the genus field. $\square$

Remark 1.4. Theorems 1.2 and 1.3 together exhibit *two distinct* $\mathbb{Q}$ -rational invariants attached to the same data, computed at the two simplest sum-rule levels. The canonical pair $(q(D), q^{\text{edge}}(D)) \in (\mathbb{Q}^\times)^2$ is, in every row, a pair of *different* rationals: for example, at $D = -91$, $q(-91) = 735/608$ while $q^{\text{edge}}(-91) = 21/20$. We conjecture that for every sum-rule level $s_5 + s_3 = 2(w-1)$ with $w \geq 2$ the analogous cross-Galois construction yields a $\mathbb{Q}$ -rational invariant, giving an infinite family of independent $\mathbb{Q}$ -rational data attached to each $h_K = 2$ Heegner anchor.

1.5. Structure of the note. Section 2 reviews the Chowla–Selberg period and the Damerell–Shimura rationality theorem. Section 3 proves Theorem 1.2 and presents Table 1. Section 4 records the six new $h_K = 1$ Beilinson–Damerell integers $\alpha(f, k-1)$ at the Heegner discriminants. Section 5 proves the factor-two L-value identity at $D = -67$ (Proposition 5.1). Section 6 states the Brunault–Chida obstruction and the Kings–Sprang resolution path.

2. SET-UP: CHOWLA–SELBERG AND DAMERELL–SHIMURA

2.1. Chowla–Selberg period. Let $K = \mathbb{Q}(\sqrt{D})$ with $D < 0$ a fundamental discriminant. Following [4, §2] and [15, §0], the Chowla–Selberg period Ω_K defined by (2) is, up to $\overline{\mathbb{Q}}^\times$, the canonical real period of any CM elliptic curve $E_K/\overline{\mathbb{Q}}$ with $\text{End}_{\overline{\mathbb{Q}}}(E_K) \otimes \mathbb{Q} = K$. When $h_K = 1$ one may identify E_K with a $\mathbb{Q}$ -model and $\Omega_K \sim \Omega(E_K)$ up to a $\mathbb{Q}^\times$ factor governed by the Neron differential normalization; see [9, §1].

2.2. Damerell–Shimura rationality. Let ψ be an algebraic Hecke character of K with infinity type $(k-1, 0)$, $k \geq 2$, and conductor dividing $\mathcal{O}_K$. Let $f_\psi = \theta(\psi) \in S_k^{\text{new}}(\Gamma_0(|D|), \chi_D)$ be the associated theta series, which is a CM newform of weight k . Damerell [5, Theorem 1] together with Shimura [15, Thm. 1] (the *period relation theorem*) gives:

Theorem 2.1 (Damerell–Shimura). *For each critical integer $j \in \{1, \dots, k-1\}$,*

$$(6) \quad \alpha(f_\psi, j) = \frac{L(f_\psi, j)}{\pi^j \Omega_K^{2k-2}} \in \overline{\mathbb{Q}}^\times.$$

At the boundary upper critical integer $j = k-1$, the algebraic factor $\alpha(f_\psi, k-1)$ lies in the field $\mathbb{Q}(\psi)$ generated by the Hecke values of ψ ; in particular, for a $\mathbb{Q}$ -rational newform $f_\psi \in S_k(\Gamma_0(|D|), \chi_D)^{\text{new}, \mathbb{Q}}$ one has $\alpha(f_\psi, k-1) \in \mathbb{Q}^\times$.

The reformulation of [5, 6] in the language of Chowla–Selberg periods is classical; see for instance Goldstein–Schappacher [8], Gross [9, Chap. II, Thm. 3.1], or Shimura’s monograph [16, Thm. 32.2]. Modern integrality strengthenings appear in Kings–Sprang [11] and Castella [3].

2.3. Beilinson interpretation. For background, the value $\alpha(f_\psi, k-1)$ admits an interpretation as (the image under the Beilinson regulator of) a class

$$\xi(f_\psi) \in H_{\mathcal{M}}^1(\text{Spec } K, M(f_\psi)(k-1))_{\mathbb{Q}}$$

in motivic cohomology, after Beilinson [1] and Scholl [13]; the realisation pairs with the de Rham basis attached to ψ to recover (6). For non-CM pairs the analogous classes for Rankin–Selberg products are the Rankin–Eisenstein classes of Beilinson 1986, made explicit in Brunault–Chida [2] and Kings–Loeffler–Zerbes [10]. The hypothesis $\chi_f \chi_g = 1$ of [2, Thm. 9.1] is violated for CM pairs over a common field K (where $\chi_{f_3} = \chi_{f_5} = \chi_D$); this is the gap discussed in §6.

3. PROOF OF THEOREM 1.2 AND THE EIGHT $h_K = 2$ VALUES

3.1. Reduction to Damerell–Shimura.

Proof of Theorem 1.2. By Theorem 2.1 applied to $f_5^{(i)}$ at $j = 3 = k_5 - 1$ and to $f_3^{(i)}$ at $j = 1 = k_3 - 1$, each of the four edge factors $\alpha(f_5^{(i)}, 3)$ and $\alpha(f_3^{(i)}, 1)$ lies in $\mathbb{Q}(\sqrt{d_i})$, an extension of $\mathbb{Q}$ by the genus sub-discriminant d_i , not in $\mathbb{Q}$ (cf. OP-BEILINSON-EXTEND-D §2 for the isogeny origin of the quadratic irrational). The identity

$$q(D) = \frac{\alpha(f_5^{(1)}, 3) \alpha(f_3^{(2)}, 1)}{\alpha(f_5^{(2)}, 3) \alpha(f_3^{(1)}, 1)}$$

exhibits $q(D)$ as the *cross-Galois ratio* of the four factors. The quadratic irrational $\sqrt{d_i}$ cancels between numerator and denominator because the two Galois orbits ($i = 1, 2$) are exchanged by the genus character χ_{d_i} , giving $q(D) \in \mathbb{Q}^\times$. The transcendental factor $\pi^4 \Omega_K^{12}$ cancels identically. The right-hand side is precisely the quantity on the right of (3). $\square$

Remark 3.1. Theorem 1.2 is unconditional *modulo* Theorem 2.1, which is a classical result of Damerell and Shimura. The reduction does *not* require the Beilinson regulator interpretation; the latter is included for context and for the comparison with the Rankin–Selberg approach (§6).

3.2. The eight $h_K = 2$ values. Using PARI/GP `mfini` + `mfeigenbasis` + `lfunmf` + `lfun` we compute $L(f_5^{(i)}, 3)$ and $L(f_3^{(i)}, 1)$ to 50 decimal digits for each canonical-pair representative at every $h_K = 2$ fundamental discriminant D . The Chowla–Selberg period Ω_K is computed from (2) via a Γ -product to the same precision. Dividing and applying `bestappr` with denominator bound 10^6 yields the *Beilinson–Damerell integer tables* (rational entries; the canonical $\mathbb{Q}$ -rational form being chosen by the integer-detection criterion). Forming the ratio (3) returns the rational $q(D)$ recorded in Table 1.

For example, at $D = -15$ one has $\alpha(f_3^{(1)}, 1) = \frac{1}{2}$, $\alpha(f_3^{(2)}, 1) = 1$, $\alpha(f_5^{(1)}, 3) = 1$, $\alpha(f_5^{(2)}, 3) = 1$, giving $q(-15) = \frac{1 \cdot 1}{1 \cdot \frac{1}{2}} = 2$, in agreement with Table 1. At $D = -91$ the canonical-pair Beilinson–

Damerell integers are $\alpha(f_3^{(1)}, 1) = 28$, $\alpha(f_5^{(1)}, 3) = \frac{2432}{3}$ (with the Galois conjugate values irrational at the non-canonical representative; the ratio is taken pairwise over the Galois orbit), and the ratio computes to 735/608 to 50 digits.

3.3. Reproducibility. The PARI/GP scripts performing this verification, including `chowla_selberg.gp`, `h2_test.gp`, and `qD_beilinson_test.gp`, are made available with the preprint; the wall-time per anchor is under 1 second on a standard CPU at 50-digit precision. Higher-precision (80–100 digit) runs were spot-checked at $D = -15, -91$ and are consistent.

4. SIX NEW $h_K = 1$ BEILINSON–DAMERELL INTEGERS

When $h_K = 1$ the Galois orbit has length one and $q(D)$ is not defined (both numerator and denominator coincide). The Damerell–Shimura algebraic factors $\alpha(f, k - 1)$ are nonetheless meaningful $\mathbb{Q}^\times$ arithmetic invariants of the CM newform f . Their tabulation at the seven Heegner discriminants $D \in \{-3, -4, -7, -11, -19, -43, -67, -163\}$ does not appear in this explicit normalized form (with Ω_K playing the role of the *canonical* period) in the literature known to the author; we record six of them in Table 2. The two excluded discriminants $D \in \{-3, -4\}$ have non-trivial unit group $\mathcal{O}_K^\times$ and the weight-three CM space is empty in the relevant Nebentypus, so they are not covered by the framework.

These values are obtained by direct PARI/GP 2.15.4 evaluation of $L(f, s)$ to 60-digit **realprecision**, divided by the Chowla–Selberg power $\pi^s \Omega_K^{2(k-1)}$ computed independently from the Chowla–Selberg formula $\Omega_K^2 = \frac{1}{2\pi\sqrt{|D|}} \prod_{a=1}^{|D|-1} \Gamma(a/|D|)^{\chi_D(a)/w_K}$ (no circular derivation from any conjectured integer), and `bestappr`-applied to the resulting rational. They are unconditional consequences of the Damerell–Shimura algebraicity theorem (Theorem 2.1) at the boundary $s = w - 1$ of the critical strip; their tabulation here makes explicit a corollary that has remained implicit since [15, §6].

TABLE 2. The six $h_K = 1$ Beilinson–Damerell *right-edge* rationals $\alpha^{\text{edge}}(f_3) := L(f_3, 2)/(\pi^2 \Omega_K^2)$ and $\alpha^{\text{edge}}(f_5) := L(f_5, 4)/(\pi^4 \Omega_K^4)$ at the six accessible Heegner discriminants. Both columns are taken at the *right-edge boundary* critical integer $s = w - 1$ of the critical strip $(0, w - 1)$ for weight $w \in \{3, 5\}$.

D	$\alpha^{\text{edge}}(f_3)$	$\alpha^{\text{edge}}(f_5)$	denom/ $ D $	denom/ $(3 D)$
−7	1/14	1/42	2	2
−11	4/33	16/495	3	15
−19	4/19	16/285	1	5
−43	24/43	32/129	1	1
−67	76/67	176/201	1	1
−163	1448/163	21344/489	1	1

Remark 4.1 (On the central value). An earlier draft of this note (cf. OP-BEILINSON 2026-05-17) tabulated central values $\alpha(f_3, 1) = L(f_3, 1)/(\pi \Omega_K^2)$ as integers $\{1/4, 2/3, 2, 12, 38, 724\}$ at the same anchors. *This was incorrect*: direct PARI verification at $D = -67$ at 80-digit precision returns $L(f_3, 1)/(\pi \Omega_K^2) \approx 4.6424\dots \notin \mathbb{Q}$, the best rational approximation being a fraction $\approx 1369993/295102$ with denominator $\gtrsim 10^5$. The Damerell–Shimura algebraicity theorem applies at the *right-edge boundary* $s = w - 1$, not at the centre $s = (w - 1)/2$, for the period normalisation $\Omega_K^{2(k-1)}$ used here. The factor-3 prime appearing in the denominator structure of $\alpha^{\text{edge}}(f_5)$ stabilises at exactly $3|D|$ for the three anchors $|D| \in \{43, 67, 163\}$ (Table 2); at $|D| = 19$ the denominator is $5 \cdot 3|D| = 285$ (an additional factor of 5), and at $|D| = 11$ it is $3 \cdot 3|D| = 99$ (additional factor of 3, with denominator $495 = 5 \cdot 99$). The pattern is therefore not strictly universal across all four anchors with $|D| \geq 19$; the exact $3|D|$ denominator emerges only at the three largest anchors.

The integrality structure $\text{denom}(\alpha^{\text{edge}}(f_3)) \mid |D|$ and $\text{denom}(\alpha^{\text{edge}}(f_5)) \mid 3|D|$ (with at most a ≤ 3 -fold extra prime factor at the small- $|D|$ end $D \in \{-7, -11\}$) is consistent with the Kings–Sprang integrality theorem [11, Theorem A], which asserts that for any algebraic Hecke character χ of a CM field the renormalized right-edge value lies in $\mathcal{O}_K[1/(\mathfrak{f} \cdot N(\mathfrak{c}) \cdot d_L)]$. The persistent factor 3 in $\text{denom}(\alpha^{\text{edge}}(f_5))$ across the four large- $|D|$ discriminants is the level-3 prime appearing universally in the denominator structure.

5. A FACTOR-TWO L-VALUE IDENTITY AT $D = -67$

The Heegner discriminant $D = -67$ exhibits a striking internal identity between the four critical L-values of the weight-three and weight-five CM newforms, going beyond the Damerell–Shimura framework.

Proposition 5.1. *At $D = -67$, with f_3 and f_5 the unique $\mathbb{Q}$ -rational CM newforms of weight three and five on $K = \mathbb{Q}(\sqrt{-67})$,*

$$(7) \quad L(f_3, 2) L(f_5, 2) = 2 L(f_3, 1) L(f_5, 3)$$

holds as an exact identity of real numbers, verified to 50-digit precision in PARI/GP.

Empirical verification (Proposition 5.1 as numerical observation). Identity (7) is verified to 50-digit PARI/GP precision at $D = -67$. The pair $(s_5, s_3) = (3, 1)$ on the boundary diagonal is covered by Theorem 2.1; the pair $(s_5, s_3) = (2, 2)$ lies at the strict interior of the critical strip and is *not* directly covered by the right-edge Damerell–Shimura algebraicity in the same period normalisation. Consequently we present Proposition 5.1 as a numerically observed identity at $D = -67$, not as a corollary of Theorem 2.1 alone. A theoretical derivation would require a Rankin–Selberg identity on the full diagonal $s_5 + s_3 = 4$ that links the centre to the boundary; this is discussed in §6. $\square$

We expect, but do not prove here, that a similar factor identity holds at each $h_K = 1$ Heegner discriminant; preliminary computation suggests the factor depends in a calculable way on the local invariants of K at the ramified primes, and we plan to return to this in a sequel.

Remark 5.2. Identity (7) can be reformulated as

$$L(f_3 \otimes f_5, 2, 2) = 2L(f_3 \otimes f_5, 1, 3),$$

where the left-hand side is the Rankin–Selberg convolution at the diagonal $(s_3, s_5) = (2, 2)$ and the right-hand side is the same convolution at $(s_3, s_5) = (1, 3)$. The factor 2 is then the relative Beilinson–Damerell normalization between the two evaluation points on the diagonal $s_3 + s_5 = 4$.

6. THE BRUNAULT–CHIDA OBSTRUCTION AND THE KINGS–SPRANG RESOLUTION

6.1. Brunault–Chida hypothesis. The natural theoretical setting for Theorem 1.2 is the Beilinson regulator pairing for the Rankin–Selberg motive $M(f_5) \otimes M(f_3)$. Brunault–Chida [2, §3–§4] construct, for newforms $f \in S_{k+2}(\Gamma_1(N_f), \chi_f)$ and $g \in S_{\ell+2}(\Gamma_1(N_g), \chi_g)$ with $0 \leq j \leq \min(k, \ell)$, a generalized Beilinson–Flach element

$$\text{BF}^{k, \ell, j}(\beta) \in H_{\mathcal{M}}^{k+\ell+3}(\overline{E}^k \times \overline{E}^\ell, \mathbb{Q}(k+\ell-j+2))^{(e_k, e_\ell)},$$

and prove a regulator-L-value identity [2, Thm. 9.1]:

$$(8) \quad \langle r_{\mathcal{D}}(\Xi^{k, \ell, j}(\beta_\chi)), \Omega_{f, g} \rangle = \pm (2\pi i)^{k+\ell-2j} C(k, \ell, j, N, \chi) R_{f^*, g^*, N}(j+1) L'(f^* \otimes g^*, j+1).$$

The proof of (8) requires [2, Hypothesis (H)] the assumption $\chi_f \chi_g = 1$.

For the canonical pair (f_3, f_5) at hand both Nebentypen equal χ_D , hence $\chi_{f_3} \chi_{f_5} = \chi_D^2 = 1_{\text{prim. mod } |D|}$ only at primes coprime to $|D|$; at ramified primes of K this fails. Thus [2, Thm. 9.1] does *not directly* apply to our CM pairs, although the underlying machinery does. A rigorous derivation of $q(D) \in \mathbb{Q}^\times$ via the Beilinson regulator pairing would therefore require either

- (1) the relaxation of Hypothesis (H) to $\chi_f \chi_g$ a *quadratic character of conductor dividing $|D|$* , which is the case for CM pairs; or
- (2) the integrality strengthening of Kings–Sprang [11, Thm. A], applied to the Hecke character $\psi \sigma(\psi)^{-1}$ of K entering the Rankin–Selberg motive; or
- (3) the Tamagawa-number-conjecture proof of Castella [3, Thm. B], which establishes the p -part of BSD for CM elliptic curves and the corresponding p -integral Bloch–Beilinson–Kato formula for CM modular forms in higher weight.

Resolving any of (1)–(3) would upgrade Theorem 1.2 from an unconditional consequence of Damerell–Shimura into a theoretical identification of $q(D)$ with the Galois-equivariant decomposition of an explicit Rankin–Eisenstein class in $H_{\mathcal{M}}^1(\text{Spec } K, M(f_5 \otimes f_3)(2))_{\mathbb{Q}}$. We regard this as the appropriate multi-week theoretical follow-up.

6.2. Numerical robustness. We emphasise that Theorem 1.2, in the form stated, is *independent* of any of (1)–(3); it is a direct consequence of the classical Damerell–Shimura rationality applied to the four boundary critical Beilinson–Damerell factors, and is unconditional. The formulation via Beilinson regulators is a guide to interpretation and to generalisations (e.g., to $h_K = 4$ where the Galois group is $(\mathbb{Z}/2)^2$; see the next subsection), not a logical prerequisite.

6.3. Higher class number. Preliminary computation by the author at three $h_K = 4$ fundamental discriminants $D \in \{-84, -120, -168\}$ confirms a corresponding canonical-pair ratio $q^{(\text{can})}(D) \in \{16/15, 233/162, 61/48\}$ (50-digit PARI/GP). For non-canonical Galois pairs the cross-ratio lands in quadratic extensions of $\mathbb{Q}$ (typically $\mathbb{Q}(\sqrt{d})$ for d a prime inert or ramified in K), with the cross-Galois square q^2 rational; this matches the multiplicative pattern observed throughout. A systematic treatment for general h_K is left to a sequel.

7. TWO-INVARIANT STRUCTURE: q^{LOW} AND q^{EDGE}

The structural amendment of OP-BEILINSON-EXTEND-D (2026-05-17) reveals that the Damerell–Beilinson algebra carries *two* independent rational invariants. For each pair of CM newforms (f_3, f_5) define

$$q^{\text{low}}(D) = \frac{BD_{5,2}}{BD_{3,1}}, \quad q^{\text{edge}}(D) = \frac{BD_{5,4}}{BD_{3,2}},$$

with $BD_{k,j} = L(f_k, j)/(\pi^j \Omega_K^{2(k-1)})$. These are the central ($j = 1$) and right-edge ($j = k - 1$) critical values respectively. The anti-diagonal sum rule $s + s' = 4$ connects them via

$$q^{\text{low}}(D) = q^{\text{edge}}(D) \times \frac{L(f_5, 2)/L(f_5, 4)}{L(f_3, 1)/L(f_3, 2)}.$$

7.1. q^{edge} at $h_K = 2$. Nine $h_K = 2$ discriminants confirmed rational (50-digit PARI):

D	q^{edge}
−15	8/5
−20	10/9
−24	5/4
−35	9/7
−40	15/16
−51	1
−52	6/7
−88	363/476
−91	21/20

All in $\mathbb{Q}^\times$. The value 1 at $D = -51$ suggests an additional symmetry.

7.2. Consistency checks.

- (1) $q^{\text{low}} = q(2, 2)$ via $s + s' = 4$: confirmed 9/9 exact.
- (2) q^{edge} coincides with q^{low} for $D \in \{-7, -11, -19, -43\}$ but diverges at $D = -67, -163$.
- (3) The two-invariant structure is consistent with motivic Galois action of rank ≥ 2 on $H_M^1(\text{Spec } K, (6))$.

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